

1. What is Inferential Statistics?

Answer:

Inferential statistics is a branch of statistics used to make predictions and draw conclusions about a population based on sample data. It helps researchers analyse trends, patterns, and relationships without studying the entire population.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Formula Explanation

- $\bar{x}$ = Sample Mean
- $\sum x$ = Sum of observations
- n = Number of observations

Real-World Example

A healthcare researcher studies 100 patients to estimate the average BMI of all hospital patients.

2. What is Hypothesis Testing and its Components?

Answer:

Hypothesis testing is a statistical method used to determine whether a claim about a population is true or false using sample data. It includes two components: the null hypothesis (H_0), which assumes no significant relationship exists, and the alternative hypothesis (H_1), which assumes a significant relationship exists.

Formula

$$\begin{aligned} H_0: \mu_1 &= \mu_2 \\ H_1: \mu_1 &\neq \mu_2 \end{aligned}$$

Formula Explanation

- H_0 = Null Hypothesis
- H_1 = Alternative Hypothesis
- μ_1, μ_2 = Population Means

Real-World Example

Testing whether smokers and non-smokers have the same average BMI.

3. Explain Confidence Interval and Critical Value.

Answer:

A confidence interval estimates the range in which the true population parameter lies. A critical value is used in hypothesis testing to determine whether the null hypothesis should be rejected.

Formula

$$CI = \bar{x} \pm z \left(\frac{\sigma}{\sqrt{n}} \right)$$

Formula Explanation

- CI = Confidence Interval
- $\bar{x}$ = Sample Mean
- z = Critical Value
- σ = Standard Deviation
- n = Sample Size

Real-World Example

Estimating the average blood pressure of patients with 95% confidence.

4. Define p-value.**Answer:**

The p-value is the probability of obtaining the observed result if the null hypothesis is true. It helps determine whether the test result is statistically significant.

Formula

$$p \leq 0.05$$

Formula Explanation

- p = Probability Value
- 0.05 = Significance Level

Real-World Example

If the p-value for smoking vs diabetes is 0.03, smoking significantly affects diabetes prevalence.

5. Differentiate Type I and Type II Errors.**Answer:**

A Type I error occurs when the null hypothesis is rejected even though it is true. A Type II error occurs when the null hypothesis is accepted even though it is false.

Formula

$$\alpha = P(\text{Type I Error})$$
$$\beta = P(\text{Type II Error})$$

Formula Explanation

- α = Probability of Type I Error
- β = Probability of Type II Error

Real-World Example

Type I Error: Concluding smoking causes diabetes when it does not.

Type II Error: Concluding smoking does not cause diabetes when it actually does.

6. Brief Description of z-test, t-test, Chi-square Test, and ANOVA Test.

z-test

Answer:

A z-test is used to compare sample means when the population variance is known and the sample size is large.

Formula

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Real-World Example

Testing whether the average cholesterol level differs from the national average.

t-test

Answer:

A t-test compares the means of two groups when the sample size is small or the population variance is unknown.

Formula

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Real-World Example

Comparing BMI between smokers and non-smokers.

Chi-Square Test

Answer:

The chi-square test checks whether two categorical variables are associated.

Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Real-World Example

Checking whether smoking status is associated with diabetes.

ANOVA Test

Answer:

ANOVA compares the means of three or more groups to determine whether significant differences exist among them.

Formula

$$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$$

Real-World Example

Comparing BMI across different age groups.

7. What is Covariance?

Answer:

Covariance measures how two variables change together. A positive covariance indicates that both variables increase together, while a negative covariance indicates an inverse relationship.

Formula

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Formula Explanation

- X, Y = Variables
- $\bar{X}, \bar{Y}$ = Means
- n = Number of observations

Real-World Example

Analysing whether age and BMI increase together.

8. What is Correlation?

Answer:

Correlation measures the strength and direction of the relationship between two variables. Its value ranges from -1 to +1.

Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Formula Explanation

- r = Correlation Coefficient
- x, y = Variables
- $\bar{x}, \bar{y}$ = Means

Real-World Example

Studying the relationship between blood pressure and age.